SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour
QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I _G. GOWTHAMI/192524285(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:G. Gowthami

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search

ii. Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model

iv. Compare online search with uninformed and informed search strategies

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation

ii. Explain how backtracking search works in this scenario

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency

iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

iv. Design an evaluation function and justify the factors considered

v. Recommend a strategy for balancing optimality and real-time performance

ANSWERS

1. AI-Powered Drone Delivery (Greedy Best-First Search & A*)

i. Greedy Best-First Search

Selects the path with the lowest heuristic value (estimated distance).

Very fast in finding a route.

Suitable for emergency situations requiring quick decisions.

Weakness: Ignores actual travel cost, so it may choose unsafe or blocked routes.

ii. A* Search

Uses $f(n) = g(n) + h(n)$.

$g(n)$: Actual cost from the start.

$h(n)$: Estimated cost to the destination.

Finds the shortest and safest path by considering both actual and estimated costs.

iii. Impact of Heuristic Accuracy

Accurate heuristic improves both Greedy and A* performance.

Poor heuristic causes Greedy to make incorrect decisions.

A* is less affected because it also considers the actual path cost.

iv. Effect of Dynamic Changes

Blocked roads require the search to recalculate routes.

Greedy may repeatedly choose blocked paths.

A* adapts better by updating costs and finding alternative routes.

v. Recommended Algorithm

A* is recommended because:

Produces optimal routes.

Considers safety and travel cost.

Handles changing environments better than Greedy.

Provides reliable decisions during emergencies.

2. Smart City Traffic Optimization (Hill Climbing & Genetic Algorithm)

Optimization Problem

Objective:

Minimize waiting time.

Reduce fuel consumption.

Decrease traffic congestion.

Traditional search is inefficient because:

Huge search space.

Dynamic traffic conditions.

High computational cost.

i. Hill Climbing

Starts with one signal configuration.

Continuously selects better neighboring solutions.

Stops when no better solution exists.

Limitations

Gets stuck in local maxima.

Cannot escape plateaus.

Performs poorly in complex traffic networks.

ii. Local Maxima, Plateaus and Ridges

Local Maximum: Good but not the best solution.

Plateau: Many equal-quality solutions causing no progress.

Ridge: Best path requires temporary worse moves.

iii. Simulated Annealing / Genetic Algorithm

Simulated Annealing: Occasionally accepts worse solutions to escape local maxima.

Genetic Algorithm: Uses selection, crossover, and mutation to generate better solutions and explore many possibilities.

iv. Recommended Approach

Genetic Algorithm

Highly scalable.

Adapts to changing traffic.

Produces near-optimal solutions efficiently.

3. Mars Rover (Online Search Agent)

i. Why Online Search?

Environment is unknown.

Complete map is unavailable.

Rover learns while moving.

ii. Challenges

Partial observability.

Dynamic obstacles.

Limited sensor information.

Communication delays.

iii. Internal Model

The rover:

Stores explored locations.

Updates obstacle information.

Learns safe paths.

Continuously improves navigation.

iv. Online vs Offline Search

Online Search

Offline Search

Learns during exploration

Requires complete map

Handles uncertainty

Assumes fixed environment

Suitable for Mars

Not suitable

v. Best Approach

Online Search Agent

Adapts to unknown terrain.

Makes real-time decisions.

Safely avoids hazards.

4. University Exam Timetable (Constraint Satisfaction Problem)

CSP Formulation

Variables

Exam subjects.

Domains

Available time slots and exam halls.

Constraints

No overlapping exams.

Limited halls.

Faculty availability.

Subject order.

Special accommodations.

i. Variables, Domains and Constraints

Variables represent exams, domains represent possible schedules, and constraints ensure valid timetables.

ii. Backtracking Search

Assigns one exam at a time.

Checks constraints.

If conflict occurs, returns to the previous assignment and tries another option.

iii. Constraint Propagation

Forward checking:

Removes invalid future choices.

Detects conflicts early.

Reduces unnecessary searching.

iv. Best Strategy

Backtracking + Forward Checking

Efficient.

Reduces computation.

Suitable for large universities.

5. Strategic Game AI (Minimax & Alpha-Beta Pruning)

i. Minimax Algorithm

AI assumes the opponent plays optimally.

Maximizes its own score.

Minimizes opponent advantage.

Chooses the best possible move.

ii. Computational Challenges

Very large game tree.

High memory usage.

Long computation time.

Difficult for real-time games.

iii. Alpha-Beta Pruning

Eliminates unnecessary branches.

Produces the same optimal move as Minimax.

Greatly reduces computation time.

Enables faster decisions.

iv. Evaluation Function

Factors:

Player score.

Available resources.

Army strength.

Map control.

Remaining health.

Strategic position.

v. Recommended Strategy

Minimax with Alpha-Beta Pruning

Maintains near-optimal decisions.

Reduces search time.

Suitable for real-time strategy games.

Balances accuracy and speed.

RUBRICS FOR CALCUL ATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand